Deva Anand — Project Portfolio

Curated for Snowflake Software Engineering Intern — Fall 2026

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Backend engineer (2 yrs production, Wysa NHS eTriage) and MS CS at UMass Amherst focused on systems for data, performance engineering, and AI-native engineering workflows. The four projects below were chosen for direct relevance to Snowflake's Database Query Engine, Engineering Systems, Quality & Release Engineering, and Cloud Platform tracks.

Projects

Spark ETL Performance Optimization — NYC TLC Taxi Dataset github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Databricks, SQL, Python

- Optimized a large-scale Spark ETL pipeline on the NYC TLC taxi dataset using **broadcast-join tuning, partitioning, caching, adaptive Spark settings, column pruning,** and precomputed fields; reduced shuffle volume and wall-clock runtime relative to the naive baseline.
- Profiled stage-level timings via the **Spark UI** to localize shuffle and skew bottlenecks, then iterated on physical-plan choices to validate each change against measured execution metrics.
- Documented a repeatable optimization workflow comparing naive vs. tuned runs — methodology designed to be reused across downstream analytics jobs and as a teaching reference for query-engine-style performance work.

AgenticSearch — Provenance-First Entity Discovery Service github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Designed a multi-stage agentic web-discovery service that converts open-ended topic queries into structured entity tables: **typed retrieval planning** → **web search** → **page rerank** → **evidence-regime classification** → **deterministic extraction with LLM fallback** → **output verification**.
- Returns **cell-level provenance** (value, snippet, source URL/title, confidence) for every extracted result — making downstream review and audit possible without re-running the pipeline.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, and pipeline instrumentation; treated AI as a high-trust collaborator while keeping system behavior grounded in deterministic validation.
- Built a reviewer-facing trust UI for inspecting extraction quality and ranking transparency at the row level.

KG2RAG-Enhanced — Multi-Hop QA on HotpotQA github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama, sentence-transformers, llama-index, spaCy, PyTorch | UMass CS 685 Advanced NLP

- Extended KG2RAG with **multi-view seed retrieval**: generated sub-questions for each multi-hop query, retrieved passages per view, then fused via **Reciprocal Rank Fusion (RRF), cross-encoder reranking,** and **Maximal Marginal Relevance (MMR)**.
- Replaced the heuristic top-M selection with a **0–1 knapsack token-budget formulation** (value = relevance × coverage), turning evidence selection into an optimization problem under fixed context-window constraints.
- Ran **100–500 question batch experiments** via an Ollama + LLaMA-3 inference pipeline with token and latency logging to compare retrieval variants and ranking stages quantitatively.

cf-ai-research-scout — Edge-Native AI Research Assistant github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built a cloud-native AI research assistant end-to-end on Cloudflare Workers: agent fetches web sources, generates multi-step research digests, and **streams responses to the client over WebSocket** in real time.
- Used **AIChatAgent + Durable Objects** for stateful chat, **WorkflowEntrypoint** for durable multi-step workflows, **D1** for persistence, and Workers AI for embeddings and streaming inference — no external database required.
- Designed durable workflow boundaries so multi-step research tasks survive process restarts and individual tool failures; agent state persists server-side across reconnects.

Additional Repos

- **AutoEval** — retrieval-evaluation improvement agent with anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection); accepted updates are committed back to Git with report cards. github.com/Deva-1903/AutoEval
- **CS 689 Generative Models (JAX)** — RealNVP normalizing flows and DDPM diffusion implemented from scratch; reverse-mode autodiff over matrix-valued ops; 8 deep architectures benchmarked on CIFAR-10. github.com/Deva-1903/UMASS_CICS_689
- **Sonare** — offline sign ↔ speech cross-platform desktop app (Electron / React / FastAPI / MediaPipe / whisper.cpp); Qualcomm Edge AI Hackathon 2025. github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare